Multi-firm Entrepreneurship and Financial Frictions

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Research Exchange

February 16, 2017

Relationship between Wealth, Capital and Financial Frictions

- Financial frictions as collateral constraint:

$$k \leq \lambda a$$

- Suppose entrepreneur wants to borrow k^* but $k \leq \lambda a < k^*$
- λ represents degree of financial frictions
- If $\lambda = \infty \rightarrow$ perfect credit market
- If $\lambda = 1 \rightarrow$ worst credit market
- Entrepreneur may overtime accumulate wealth and <u>overcome financial frictions</u> such that $k^* < \lambda a$

Motivation

1. The negative effect of financial frictions on aggregate productivity can be reduced through wealth accumulation.
2. Current literature quantifies this effect using a crucial assumption that an entrepreneur owns only *one firm*. Loss of 20% - 30% in aggregate productivity.
3. But there is another margin: **entrepreneurs with multiple firms**.

What is the effect of financial frictions on aggregate productivity when there are multi-firm entrepreneurs?

This paper's contributions

1. **Empirical contribution**
   - Construct and introduce a new firm-entrepreneur level dataset
     - Financial information on *all* the registered firms in Thailand
     - Detailed industry classification from agriculture, manufacturing and services
     - Identify each entrepreneur and his firms

2. **Theoretical contribution**
   - Build a tractable model in which entrepreneurs endogenously choose the number of firms
   - Quantify the effect of productivity loss when there are multi-firm entrepreneurs

Findings

1. **Empirical findings**
   - Document new facts on multi-firm entrepreneurship
     - 35% of entrepreneurs have multiple firms
     - Firms owned by multi-firm entrepreneurs have lower exit rates and higher growth rates
     - Entrepreneurs with multiple firms accumulate wealth faster

2. **Theoretical preview**
   - Productivity loss due to financial frictions is 21%, but it is only 7% in a model with multi-firm entrepreneurship

Key mechanism of the model

1. **Scale**: Given DRS technology, entrepreneurs earn more by spreading his wealth across firms

   E.g.: Production function $y = \sqrt{k}$.
   Entrepreneurs with 2 units of capital k produces:
   $\sqrt{2}$ with one firm, and
   $\sqrt{1} + \sqrt{1} = 2$ with two firms with evenly split capital.

2. **Diversification**: If firms that an entrepreneur owns are not perfectly correlated, then the entrepreneur's total income will be less volatile

$1 + 2 \Rightarrow$ allow talented entrepreneurs to:

$$\left.\begin{array}{l} -\text{Earn more} \\ -\text{Exit less} \end{array}\right\} \text{Accumulate more wealth faster}$$

Outline

Data

- Description
- Fact 1: 1/3 of entrepreneurs own multiple firms
- Fact 2: Firms of multi-firm entrepreneurs grow faster, exit less
- Fact 3: Multi-firm entrepreneurs accumulate more wealth

Model

- Environment
- Recursive problem
- Getting tractability
- Calibration results

Conclusion

Data

Data

Why Thailand?

- Names of entrepreneurs and firms
- Last name of the entrepreneurs are unique Law
- Panel data of entrepreneurs that track number of firms owned over time
- All registered firms in Thailand since 1999
- Focus on domestic-owned and non-public firms
- Other features of the data:
  1. Both manufacturing and services
  2. Annual balance sheet and income statement from 2007-2015 Variables
  3. Observe years of reported financial info
  4. Firm ownership
  5. Contain small and private firms

Definition of entrepreneurs, their wealth and firms

- Entrepreneur is defined at the family level
- But firms have multiple entrepreneurs with different last names
- Classify firms belonging to the an entrepreneur if the entrepreneur owns more than 50%
- This guarantees that a firm is matched to at most one entrepreneur
- Entrepreneurs' total wealth = sum of equity from all firms

Firms' summary statistics

Variable	Firms owned by Multi-Firm Entrepreneurs	Firms owned by Single-Firm Entrepreneurs	Mean of Firms (multi) – Mean of Firms (single)
Revenue ($'000s)	1,899	701	1,198***
	(6,015)	(2,578)	
Capital ($'000s)	389	97	293***
	(1,782)	(594)	
$\frac{\text{Capital}}{\text{Equity}}$	1.15	0.753	0.402***
	(6.05)	(4.62)	
Return on Assets	0.087	0.093	-0.006***
	(0.201)	(0.231)	
Age (years)	13.01	10.47	2.54***
	(10.17)	(8.55)	
No. of Entrepreneurs	1.40	1.34	0.065***
(family)	(0.857)	(0.664)	
Total Share Owned	0.914	0.923	-0.009***
by Majority	(0.163)	(0.150)	
No. of Firms	142,455	71,889	

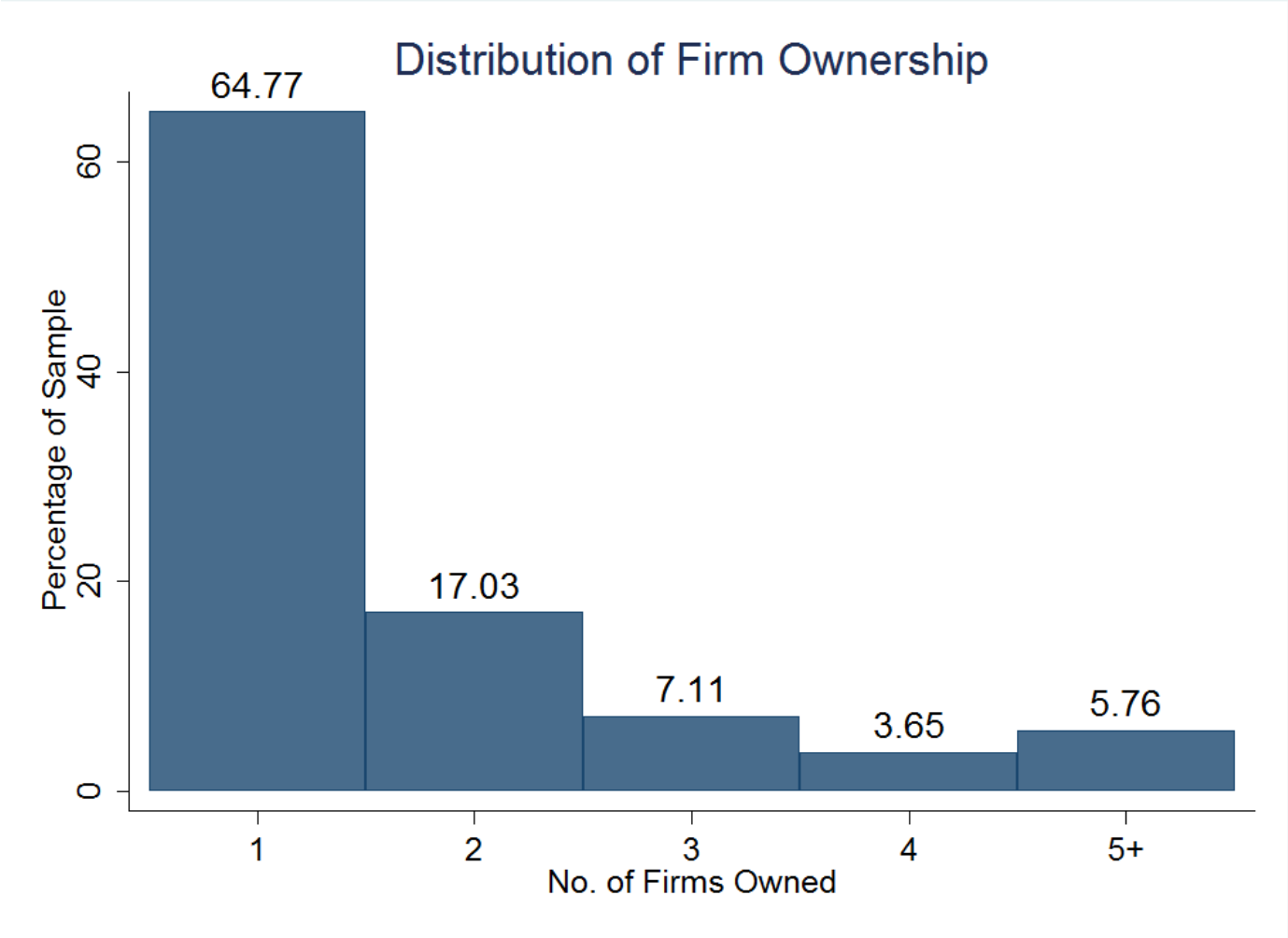
Standard deviations are in parentheses.
*** denotes significance at the 1% level.

Entrepreneurs' summary statistics

Variable	Multi-Firm Entrepreneurs	Single-Firm Entrepreneurs	Mean of Multi-Firm Ent. – Mean of Single-Firm Ent.
Wealth ($'000s)	927 (4,976)	194 (1,034)	733***
Wealth Growth Rate	0.143 (0.241)	0.094 (0.211)	0.048***
Experience (years)	21.89 (11.64)	14.57 (10.11)	7.32***
Prob. Entrepreneur Survives Subsequent Year	0.998 (0.044)	0.969 (0.175)	0.030***
No. of Firms Owned by Entrepreneur	2.62 (1.17)	1 (.)	1.62***
No. of Family Members	6.35 (3.99)	2.90 (2.32)	3.45***
No. of Entrepreneurs	40,724	71,889	

Standard deviations are in parentheses.
*** denotes significance at the 1% level.

Fact 1: 1/3 of entrepreneurs own multiple firms



Data as of 2015

Composition of entrepreneurs' firms

No. of Firms Owned	Obs.	Share of entrepreneurs with firms in j unique sectors 1	2	3	4	5+
1	113,926	100.00				
2	29,961	40.15	59.85			
3	12,512	20.39	52.2	27.41		
4	6,426	11.62	39.79	38.17	10.41	
5+	13,070	4.41	17.66	30.91	24.84	12.26

Data as of 2015

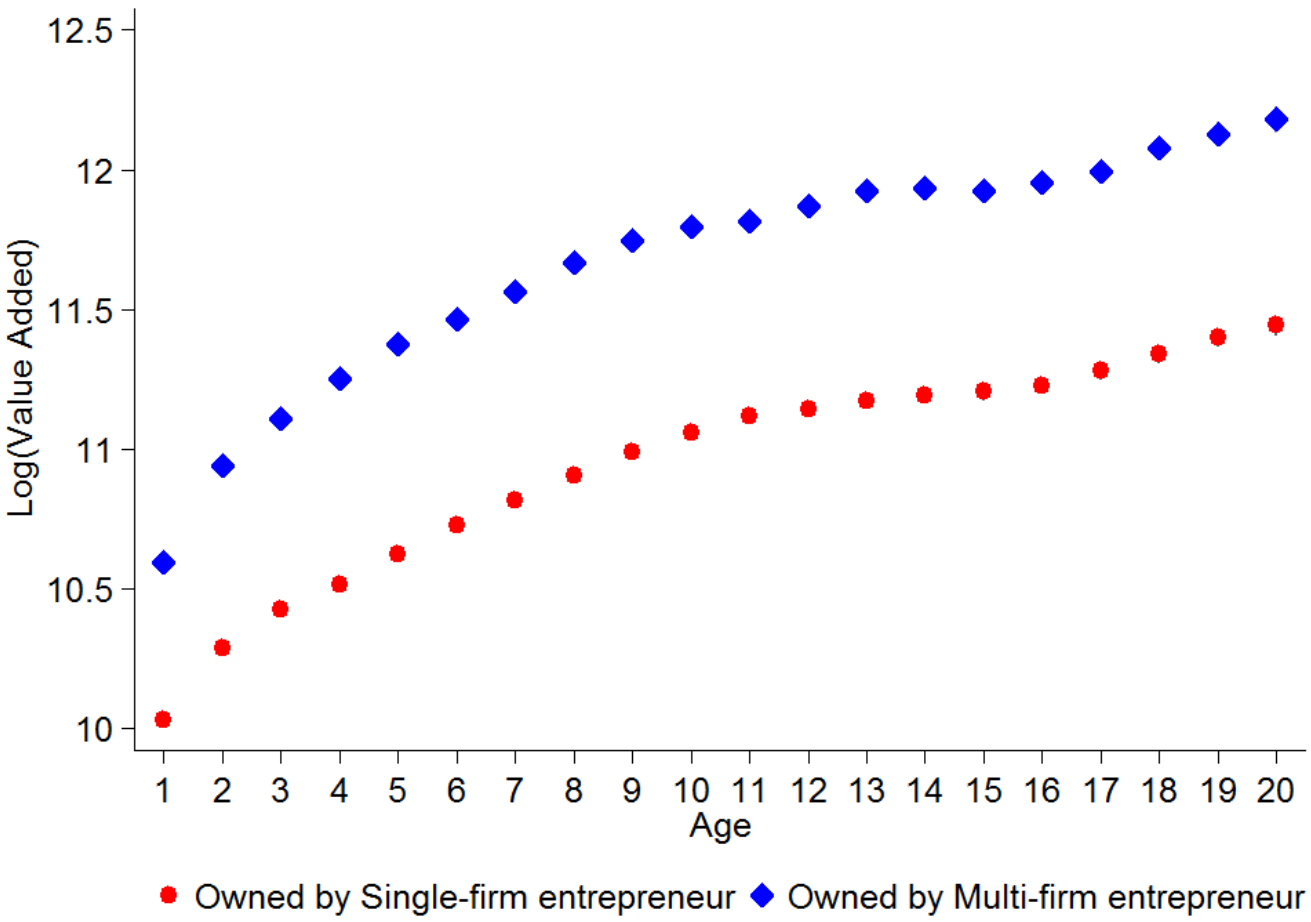
Multi-firm entrepreneurs expand to other sectors

Entrepreneurs who started in Wholesale (51) expanded into these sectors:

ISIC code	Sector	Percent
51	Wholesale trade, exl. Motor vehicles	27.64
52	Retail trade, exl. Motor vehicles	16.71
50	Sale of motor vehicles and motorcycles	3.93
55	Hotels and restaurants	2.28
70	Real estate activities	7.87
45	Construction	5.35
15	Manufacturing of Food/Beverages	2.55
63	Auxilliary transport activities	2.30
60	Land transport	2.23
29	Manufacture of Machinery and equipment	2.01
	Others	27.13

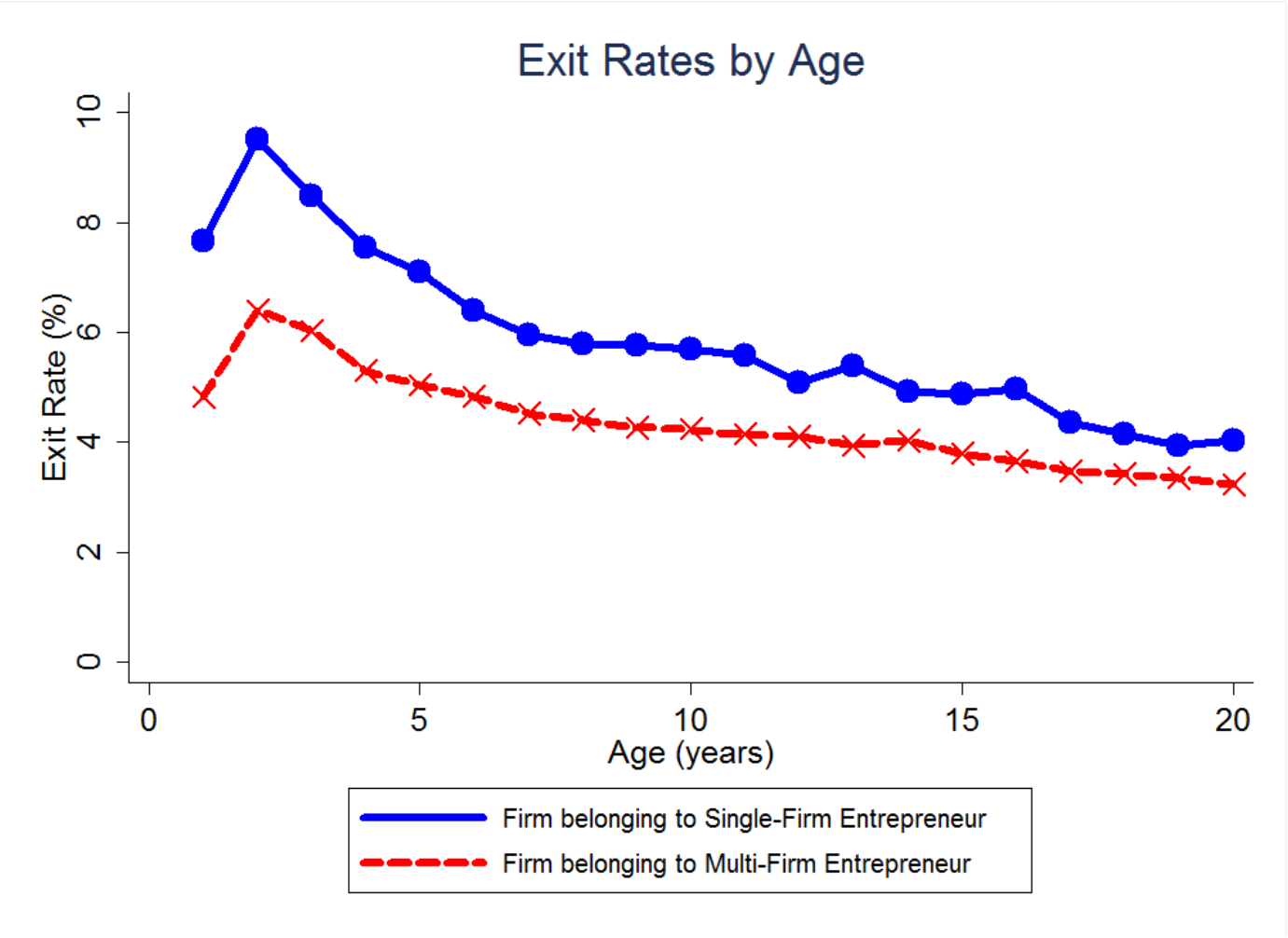
Share of entrepreneurs with unique sectors
Share of entrepreneurs with multiple firms

Fact 2a: Firms owned by multi-firm entrepreneurs are larger and grow faster



Growth rate with controls

Fact 2b: Firms owned by multi-firm entrepreneurs survive longer



Survival function based on entrepreneur's degree of diversification
Exit rate with controls

Measuring the entrepreneur's portfolio

- For each firm i owned by entrepreneur j at time t:

$$RoE_{ijt} = \frac{\text{Profit}_{ijt}}{\text{Equity}_{ijt}}$$

- He spends a share s_{ijt} of his wealth on firm i:

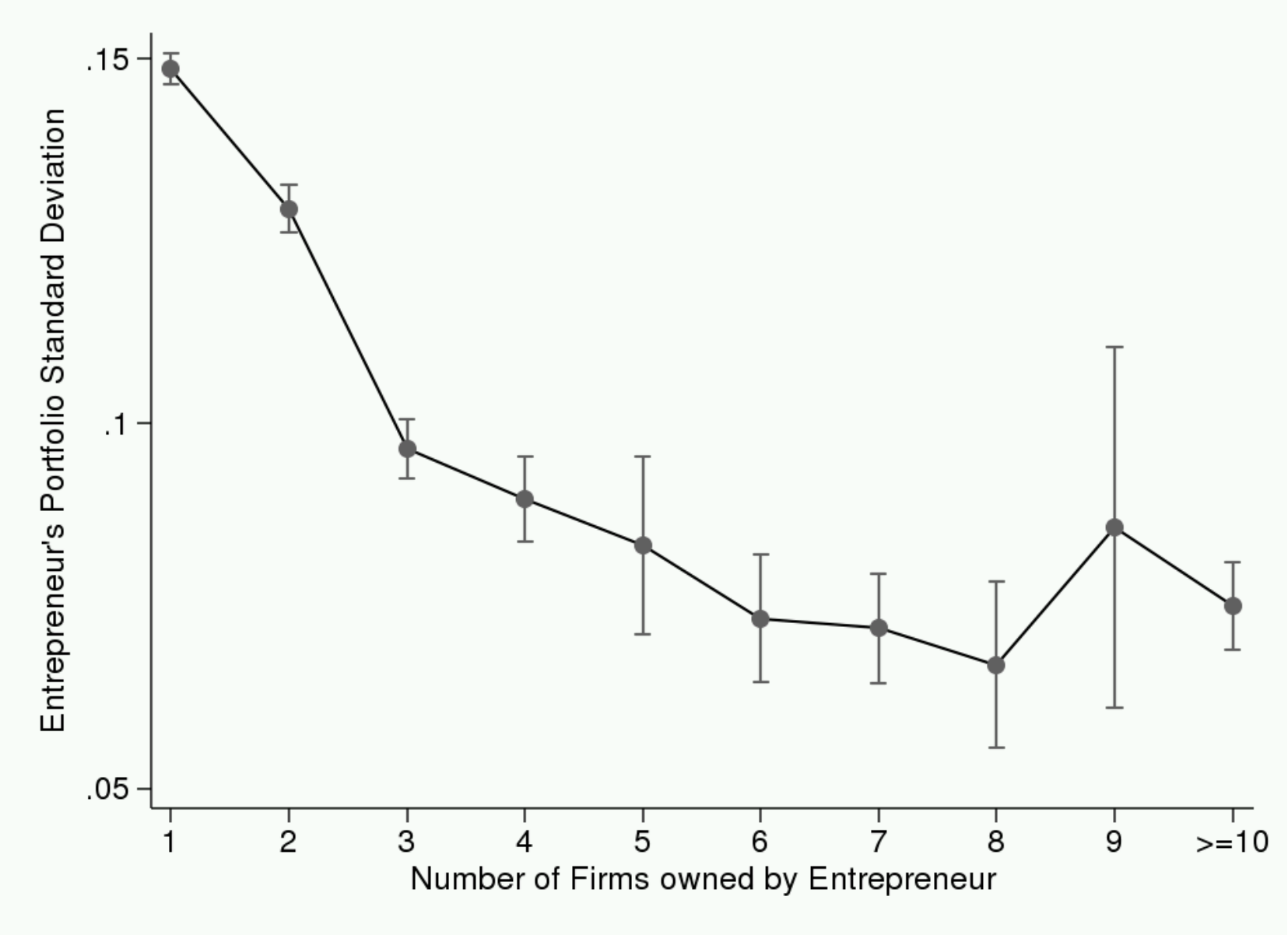
$$s_{ijt} = \frac{Equity_{ijt}}{\sum_{i=1}^{N} \text{Equity}_{ijt}}$$

- Entrepreneur return on his portfolio of N firm is then

$$\text{RoE}_{jt} = \sum_{i=1}^{N} s_{ijt} RoE_{ijt}$$

- For each entrepreneur j, calculate variance of RoE over time.

Standard deviation of entrepreneur's return and number of firms



Firms exit less as entrepreneurs diversify more

	(1) P(Firm Exit)	(2) P(Firm Exit)	(3) P(Firm Exit)	(4) P(Firm Exit)
Entrepreneur's Portfolio Volatility	0.457***	0.313***	0.295***	
	(5.00)	(3.28)	(3.21)	
Entrepreneur's 1-Herfindahl				-0.181***
				(-2.85)
Firm RoE		-0.106*	-0.114**	-0.171***
		(-1.92)	(-2.17)	(-3.29)
Firm Asset		-0.0872***	-0.108***	-0.114***
		(-6.96)	(-7.99)	(-8.25)
Other Controls				
Firm Age & Cohort	N	Y	Y	Y
Firm Industry	N	Y	Y	Y
Entrepreneur Exp& Cohort	N	N	Y	Y
Entrepreneur First Industry	N	N	Y	Y

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Addressing unobserved productivity and financial frictions

- Follow same specification as Gopinath et al (2015):
- In a frictionless world, net worth A of the firm does not affect how much a firm can accumulate capital k:

$$\frac{k_{2014} - k_{2007}}{k_{2007}} = d_{2007} + \alpha \log(A_{2007}) + \gamma \log(k_{2007}) + \beta \log(Z_{2007}) + u_t$$

- In the null hypothesis, α should be 0
- Need a measure of firm's productivity
- Solution: merge with manufacturing survey in 2007 which has labor and capital at the establishment level

Initial wealth affects capital growth

$$\frac{k_{2014} - k_{2007}}{k_{2007}} = d_{2007} + \alpha \log(A_{2007}) + \gamma \log(k_{2007}) + \beta \log(Z_{2007}) + u_t$$

	(1) $\frac{k_{2014} - k_{2007}}{k_{2007}}$	(2) $\frac{k_{2014} - k_{2007}}{k_{2007}}$	(3) $\frac{k_{2014} - k_{2007}}{k_{2007}}$
Ln(Equity)	0.204***	0.189***	
	(0.04)	(0.04)	
Ln(Productivity)	0.181***	0.238***	
	(0.04)	(0.04)	
Total No. of Firms owned by Entrepreneur			
Ln(Capital)	-0.359***	-0.393***	
	(0.04)	(0.04)	
Firm-Year Obs.	5193	5193	
Sector FE	N	Y	
Geographical Location FE	N	Y	

standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Effect on debt

...and number of firms also affects capital growth!

$$\frac{k_{2014} - k_{2007}}{k_{2007}} = d_{2007} + \alpha \log(A_{2007}) + \gamma \log(k_{2007}) + \beta \log(Z_{2007}) + u_t$$

	(1) $\frac{k_{2014} - k_{2007}}{k_{2007}}$	(2) $\frac{k_{2014} - k_{2007}}{k_{2007}}$	(3) $\frac{k_{2014} - k_{2007}}{k_{2007}}$
Ln(Equity)	0.204***	0.189***	0.167***
	(0.04)	(0.04)	(0.05)
Ln(Productivity)	0.181***	0.238***	0.277***
	(0.04)	(0.04)	(0.05)
Total No. of Firms owned by Entrepreneur			0.021**
			(0.01)
Ln(Capital)	-0.359***	-0.393***	-0.423***
	(0.04)	(0.04)	(0.04)
Firm-Year Obs.	5193	5193	3608
Sector FE	N	Y	Y
Geographical Location FE	N	Y	Y

standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Fact 3: Multi-firm entrepreneurs accumulate more wealth

Full sample period

	(1) $\frac{W_t - W_{t-1}}{W_{t-1}}$	(2) $\frac{W_t - W_{t-1}}{W_{t-1}}$	(3) $\frac{W_t - W_{t-1}}{W_{t-1}}$
Ln(Wealth)	-0.003***	-0.005***	-0.009***
	(0.0004)	(0.0004)	(0.0008)
Total No. of Firms Owned by Entrepreneur	0.009***	0.012***	0.007***
	(0.0005)	(0.0005)	(0.0006)
Experience	-0.001***	-0.003***	-0.003***
	(0.0001)	(0.0001)	(0.0002)
Experience^2	4.31×10^{-6}	0.00003***	-0.00002***
	(2.23×10^{-6})	(2.27×10^{-6})	(3.93×10^{-6})
Entrepreneur-Year Obs.	277,495	277,495	61,430
R^2	0.0038	0.0297	0.0316
Year FE	N	Y	Y
First Sector FE	N	Y	Y
Sector Interaction FE	N	N	Y

Standard errors are in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Model

Key mechanism of the model

1. **Scale**: Given DRS technology, entrepreneurs earn more by spreading his wealth across firms
2. **Reduces Volatility**: If each firm that the entrepreneurs own are not perfectly correlated, then entrepreneur's total income will be less volatile

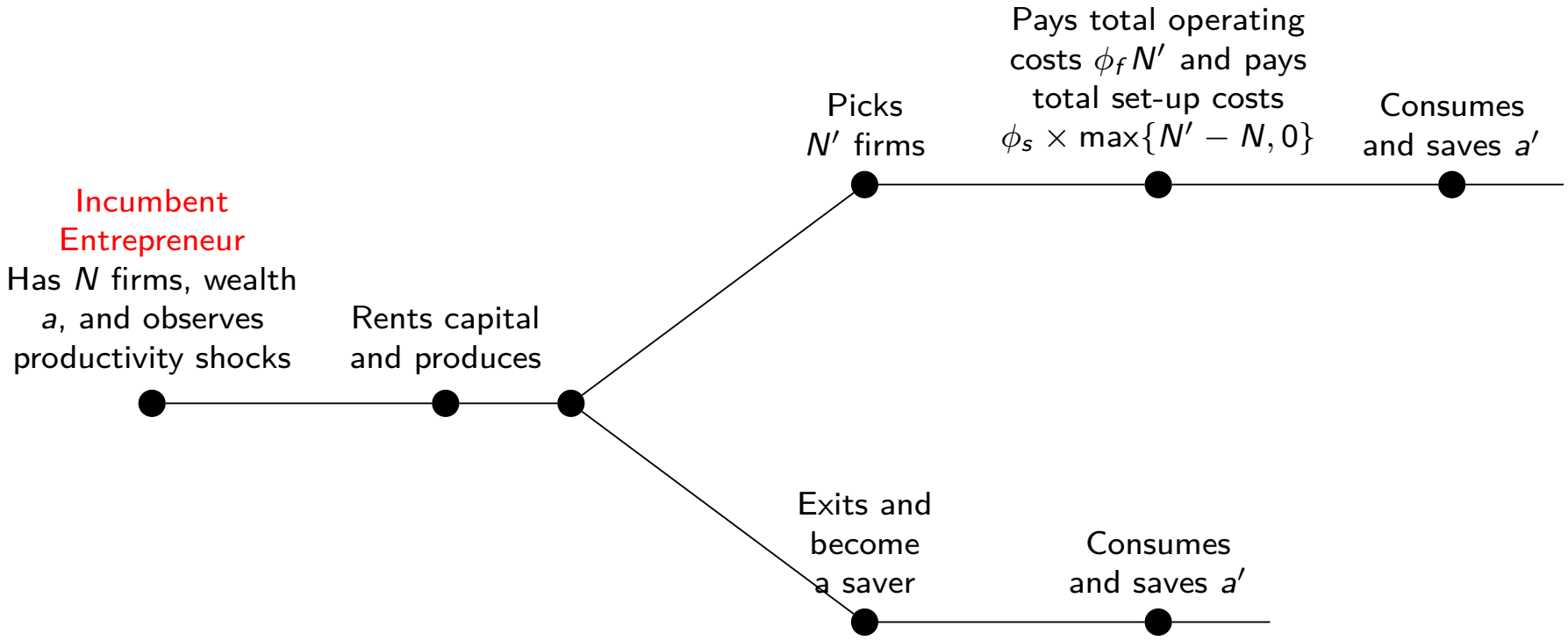
$1 + 2 \Rightarrow$ allows talented entrepreneurs to

$$\left.\begin{array}{l} -\text{Accumulate more wealth} \\ -\text{Exit less} \end{array}\right\} \text{Overcome financial frictions}$$

Key elements

- Entrepreneurs, potential entrants, and savers
- Financial frictions in the form of collateral constraint
- Wealth accumulation
- Entrepreneurs decide how many firms they want to operate
- Tractable
- Costs associated in operating each firm ϕ_f
- One time set up cost of ϕ_s for each additional firm.

Timing in period t for incumbent entrepreneurs



Production function

Continuum of entrepreneurs: wealth a, productivity z, N and $\vec{e}$.

$$Y_j = [\ \underbrace{\exp(z)}_{\substack{\text{Entrepreneur's}\\ \text{productivity}}}\ \underbrace{\exp(e_j)}_{\substack{\text{Firm } j\\ \text{productivity}}}\ ]^{1-\alpha}(k_j)^{\alpha} \tag{1}$$

$$\vec{e} \sim \mathcal{N}(\vec{0}, \Sigma) \quad \text{where} \quad \text{var}(e_j) = \sigma_e^2, \quad \text{corr}(e_j, e_i) = \rho$$

$$z' = \rho_z z + \sigma_z \epsilon_z$$

Entrepreneur's profit

- Entrepreneur's total income

$$Y = \sum_{j=1}^{N} Y_j = \sum_{j=1}^{N} \exp(z + e_j)^{1-\alpha}(k_j)^{\alpha} \tag{2}$$

- Entrepreneur rents capital k_j for firm y_j at rate R.

$$\tilde{\pi}(a, z, \vec{e}, N) = \sum_{j=1}^{N} \exp(z + e_j)^{1-\alpha}(k_j)^{\alpha} - R \sum_{j=1}^{N} k_j \tag{3}$$

Recursive problem: incumbent entrepreneur

$$V(a, z, \vec{e}, N) = \max_{\substack{a', c, \\ N', \{k_j\}_{j=1}^{N}}} \begin{cases} u(c) + \beta \int_{\hat{e}'} \int_{z'} V(a', z', \vec{e}', N') dH(z'|z) dF(\vec{e}'), & \text{if } N' > 0 \\ \\ u(c) + \beta V_s(a'), & \text{if } N' = 0 \end{cases}$$

subject to

$$c + a' = \tilde{\pi}(a, z, \vec{e}, N) - N'\phi_f - \phi_s \max\{N' - N, 0\} + (1 + r)a$$

and the borrowing constraint:

$$\sum_{j=1}^{N} k_j \leq \lambda a$$

Entrepreneur's profit maximization

$$\tilde{\pi}^*(a, z, \vec{e}, N) = \max_{\{k_j\}_{j=1}^N} \left\{ \sum_{j=1}^N \exp(z + e_j)^{1-\alpha} (k_j)^\alpha - R \sum_{j=1}^N k_j \right\} \quad (4)$$

$$s.t. \qquad \sum_{j=1}^N k_j \leq \lambda a \quad (5)$$

Optimal allocation of static profit

$$\tilde{\pi}^*(z, a, \vec{e}, N) = \begin{cases} (1-\alpha)\left[\left(\frac{\alpha}{R}\right)^{\alpha}\right]^{\frac{1}{1-\alpha}} \exp(z\ )\sum_{j=1}^{N} \exp(e_j), & \text{if unconstrained} \\ (\lambda a)^{\alpha}\left[\exp(z\ )\sum_{j=1}^{N} \exp(e_j)\right]^{1-\alpha} - R\lambda a, & \text{if constrained} \end{cases} \tag{6}$$

constrained if

$$\left[\left(\frac{\alpha}{R}\right)^{\alpha}\right]^{\frac{1}{1-\alpha}} \exp(z\ )\sum_{j=1}^{N} \exp(e_j) \geq \lambda a \tag{7}$$

$e \equiv \sum_{j=1}^{N} \exp(e_j)$ is a **sufficient statistics** for $\vec{e}$.

$$\tilde{\pi}^*(z, a, \vec{e}, N) = \pi^*(z, a, e, N)$$

Approximating the distribution of e

Rewriting the recursive problem for the incumbents

$$V(a, z, e, N) = \max_{a', c, N'} \begin{cases} u(c) + \beta \int_{e'} \int_{z'} V(a', z', e', N') dH(z'|z) dF(e'), & \text{if } N' > 0 \\ u(c) + \beta V_s(a'), & \text{if } N' = 0 \end{cases}$$

subject to

$$c + a' = \pi^*(a, z, e, N) - N'\phi_f - \phi_s \max\{N' - N, 0\} + (1 + r)a$$

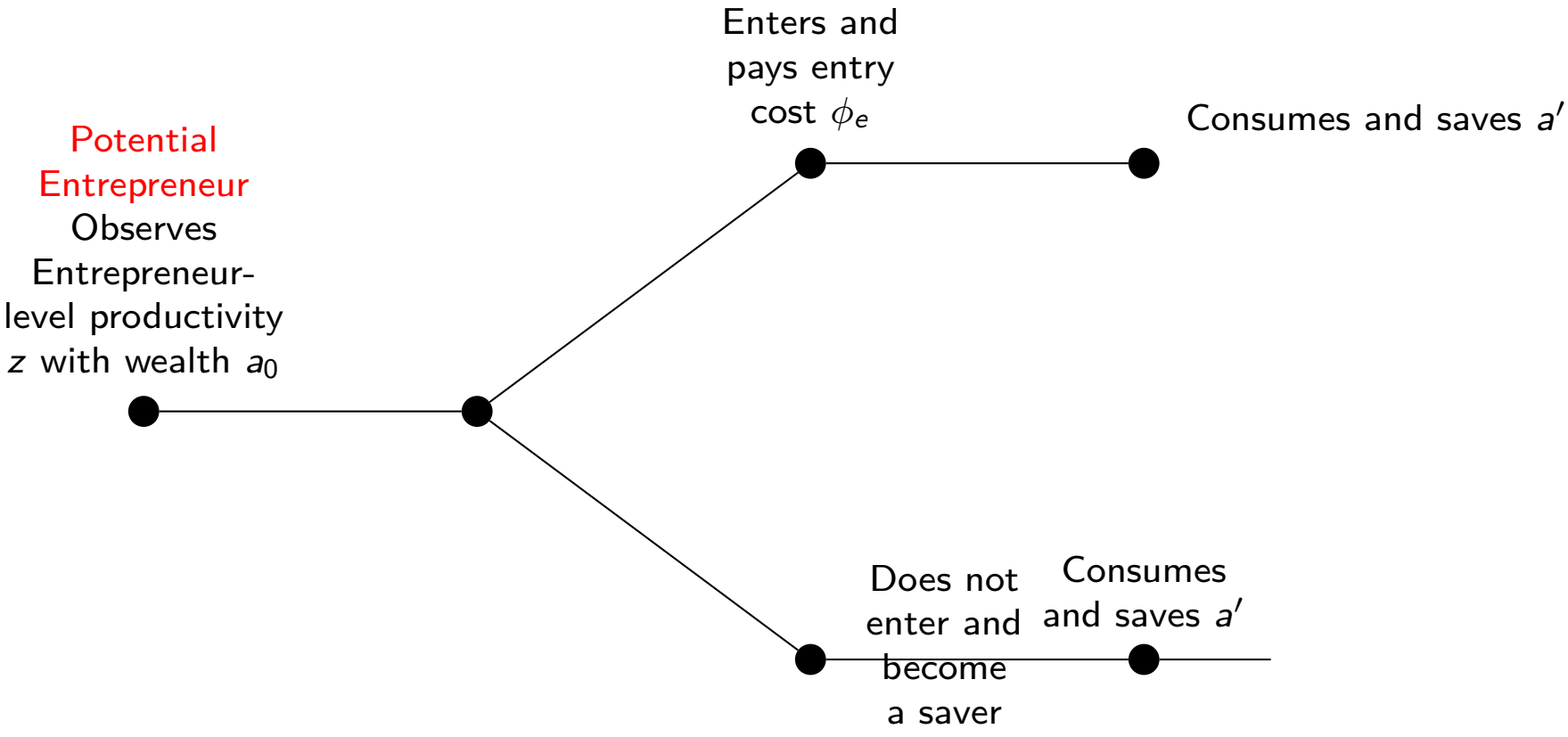
Value function of savers

$$V_s(a) = \max_{a',c} \left\{ u(c) + \beta V_s(a') \right\}$$

subject to

$$c + a' = (1 + r)a$$

Timing in period t for potential entrants



Recursive problem: potential entrepreneur

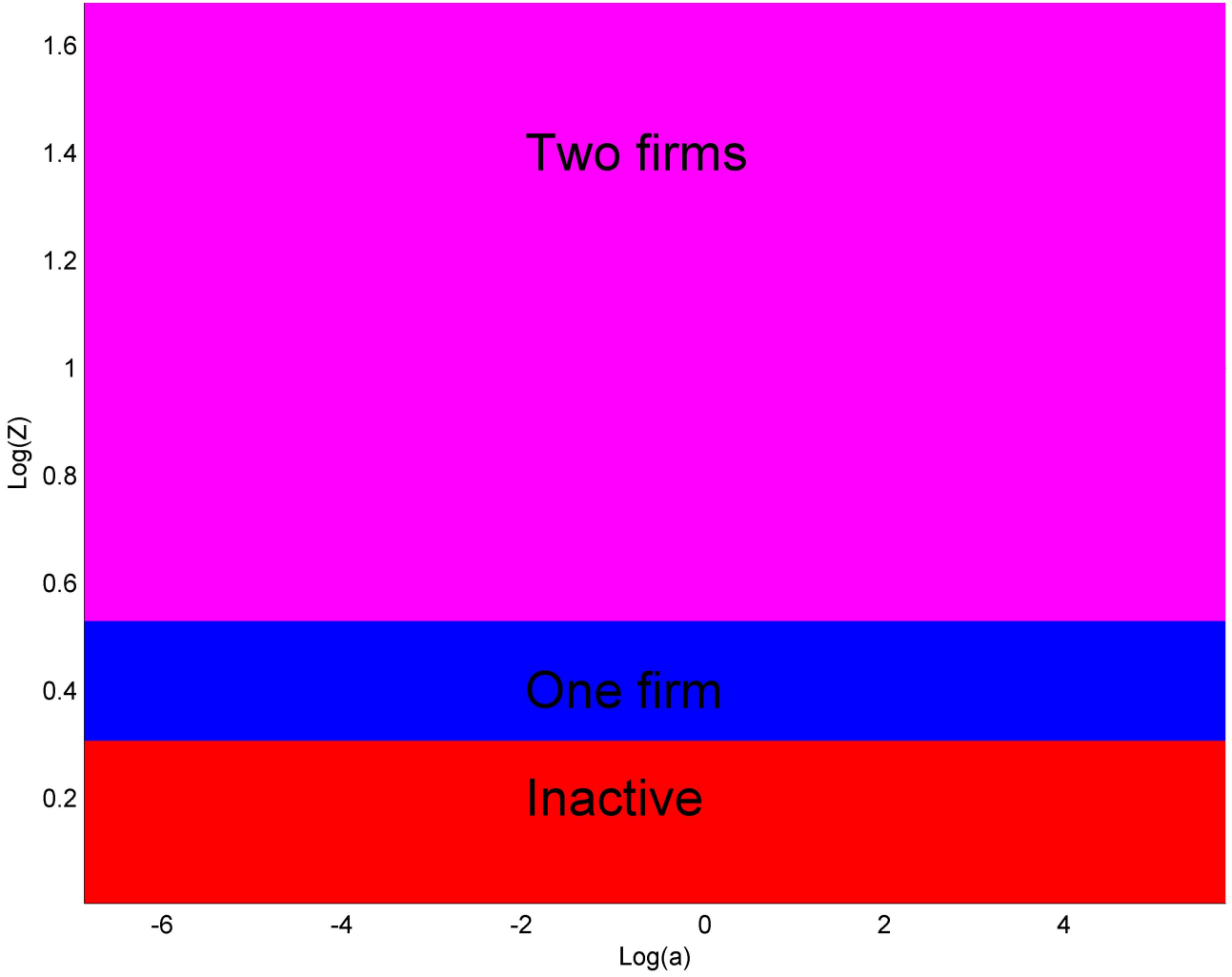
The bellman equation of a prospective entrant with wealth a_0 and productivity z is

$$V_e(a_0, z) = \max \begin{cases} \max_{a',c} u(c) + \beta V_s(a') & \text{,saver} \\ \max_{a',c} u(c) + \beta \int_{e'} \int_{z'} V(a', z', e', 1) dF(z'|z) dH(e') & \text{,enters} \end{cases}$$

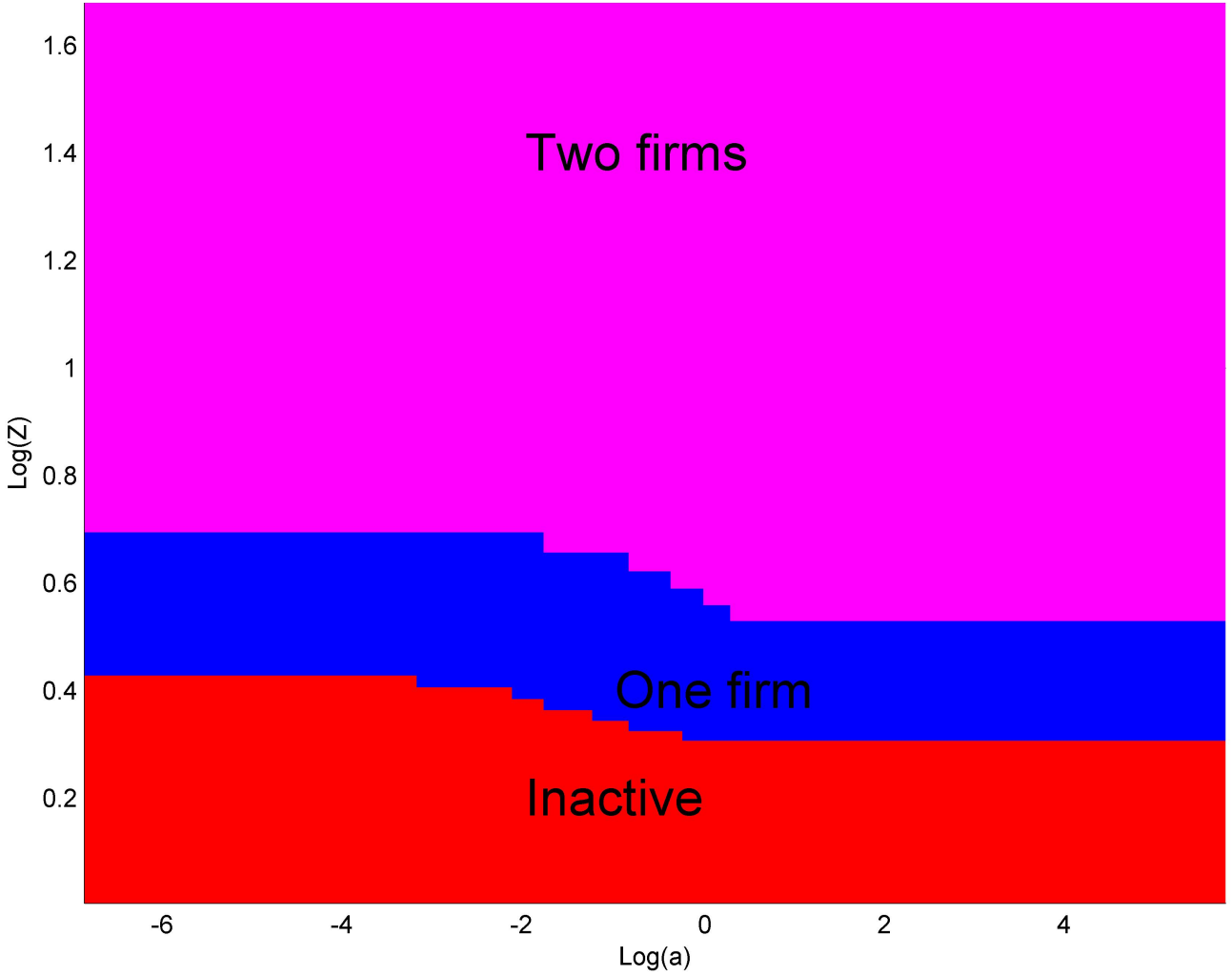
subject to

$$c + a' = (1 + r)a_0 - \phi_e \mathbf{I}(\text{enter})$$

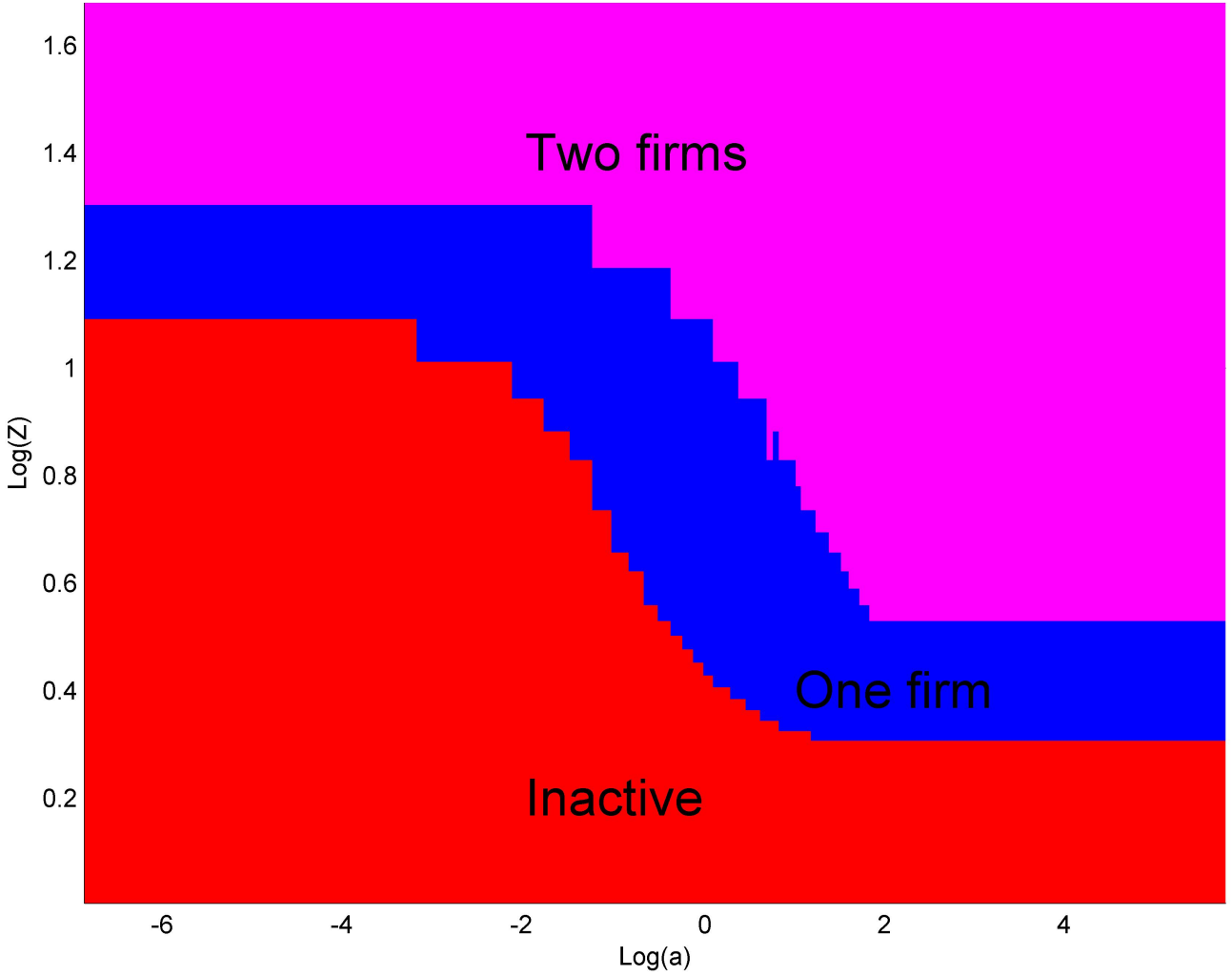
Heatmap of entrepreneurs: Perfect credit market



Heatmap of entrepreneurs: Minor imperfect credit market



Heatmap of entrepreneurs: Major imperfect credit market



Stationary Competitive Equilibrium

Let $\mu_t^n(a, z, \hat{e})$ be the measure of entrepreneurs with firms n. Given input prices, a stationary competitive equilibrium is an invariant distribution of wealth and entrepreneurial productivity and policy functions $c(a, z, \hat{e}, N)$, $a'(a, z, \hat{e}, N)$, $k(a, z, \hat{e}, N)$ and $N'(a, z, \hat{e}, N)$ such that:

1. $V_e(a_0, z)$ solves the entrant's problem.
2. $V_s(a)$ solves the saver's problem.
3. $V(a, z, \hat{e}, N)$ solve the incumbent's problem.
4. Joint distribution of wealth and productivity is a fixed for each entrepreneur with n firms.

Calibration strategy

- Benchmark model includes labor
- Span-of-control parameter (η) in the production function.
- The model simulates moments that can be matched to the data
- Choose parameters jointly such that the simulated moments match the empirical moments
- Calculate inefficiency losses based on the chosen parameters.

Assigned Parameters	Symbol	Value
Discount factor	β	0.92
Relative Risk aversion	γ	1.5
Capital share	$1-\alpha$	0.33
Span of control	η	0.85
Depreciation rate	δ	0.06
Interest Rate	r	0.04
Wage Rate	w	1

Calibrated Parameters	Symbol	Value
Strength of rental market	λ	1.37
Persistence on entrepreneur-level productivity	ρ_z	0.982
Variance(entrepreneur-level productivity shock)	σ_z^2	0.36
Variance(firm-level productivity shock)	σ_e^2	0.11
Pair-wise correlation, firm-level productivity shock	ρ_e	-0.47
Firm per-period fixed cost	ϕ_f	0.12
Additional firm set-up cost	ϕ_s	1.5
Scale parameter of Pareto distribution	ν	7.5
Initial wealth of entrant	a_0	0.66

	Thai Data	Model
SD $\log y_t$	1.63	1.60
Corr($\log y_t$,$\log y_{t-1}$)	0.85	0.81
Corr($\log y_i$,$\log y_j$)	0.68	0.68
Relative Size of Entrant	0.17	0.16
Average Leverage $\frac{k_t}{a_t}$	0.89	0.97
Share of multi-firm entrepreneur	0.35	0.38
Share of entrepreneurs, age 1-10	0.21	0.25
Exit rate of entrant entrepreneur (in percent)	5.50	5.54
Exit rate of entrepreneur (in percent)	2.28	2.29

Aggregate TFP in the benchmark model

$$Y = \frac{\left(\sum_{j=1}^{N}\left[\int_{i\in j}\exp(z_i+\hat{e}_i)(R+\theta_i)^{\frac{-(1-\alpha)\eta}{1-\eta}}di\right]\right)^{1-\alpha\eta}}{\left(\sum_{j=1}^{N}\left[\int_{i\in j}\exp(z_i+\hat{e}_i)(R+\theta_i)^{\frac{-1+\alpha\eta}{1-\eta}}di\right]\right)^{(1-\alpha)\eta}}\left(L^{\alpha}K^{1-\alpha}\right)^{\eta} \tag{8}$$

where θ_i is the shadow cost of savings for entrepreneur i.

Aggregate TFP of the social planner

Social planner solves the following problem:

$$\max_{k_i, l_i} \left(\sum_{j=1}^{N} \left[\int_{i \in j} \exp(z_i + \hat{e}_i)^{1-\eta} \left(l_i^{\alpha} k_i^{1-\alpha} \right)^{\eta} di \right] \right)$$

subject to the same K, L, and number of entrepreneurs in the benchmark economy.

$$Y_e = \left(\sum_{j=1}^{N} \left[\int_{i \in j} \exp(z_i + \hat{e}_i) di \right] \right)^{1-\eta} \left(L^{\alpha} K^{1-\alpha} \right)^{\eta} \tag{9}$$

Results

Economy	Implied parameters	TFP	Loss
Multi-firm economy, (with diversification)	Calibrated (parameters)	1.86	6.1%
Multi-firm economy (with diversification, social planner)	previous	1.98	

Results and Counterfactuals

Economy	Implied parameters	TFP	Loss
Single-firm economy	$\phi_s = \infty$	1.57	20.7%
Multi-firm economy, (no diversification)	$\rho_e = 1$	1.80	9.1%
Multi-firm economy, (with diversification)	Thai Economy	1.86	6.1%
Multi-firm economy (with diversification, social planner)	previous	1.98	

Policy Implications

- Different countries with different start-up cost ϕ_s.
- Countries with high ϕ_s will look more similarly to the one-firm economy
- Reducing start-up cost ϕ_s will lead to an increase in efficiency in that country's economy
- Can shift funds from direct lending to reducing ϕ_s

Extensions

- Add entrepreneur's time as input (Akcigit et al., 2016b)

$$y_j = (z_j)^{1-\eta}(k_j^{\alpha} l_j^{\beta} t_j^{1-\alpha-\beta})^{\eta}$$

such that $\sum t_j \leq 1$

▸ Firm exit rate and number of firms owned by entrepreneurs

- Entrepreneurs own capital

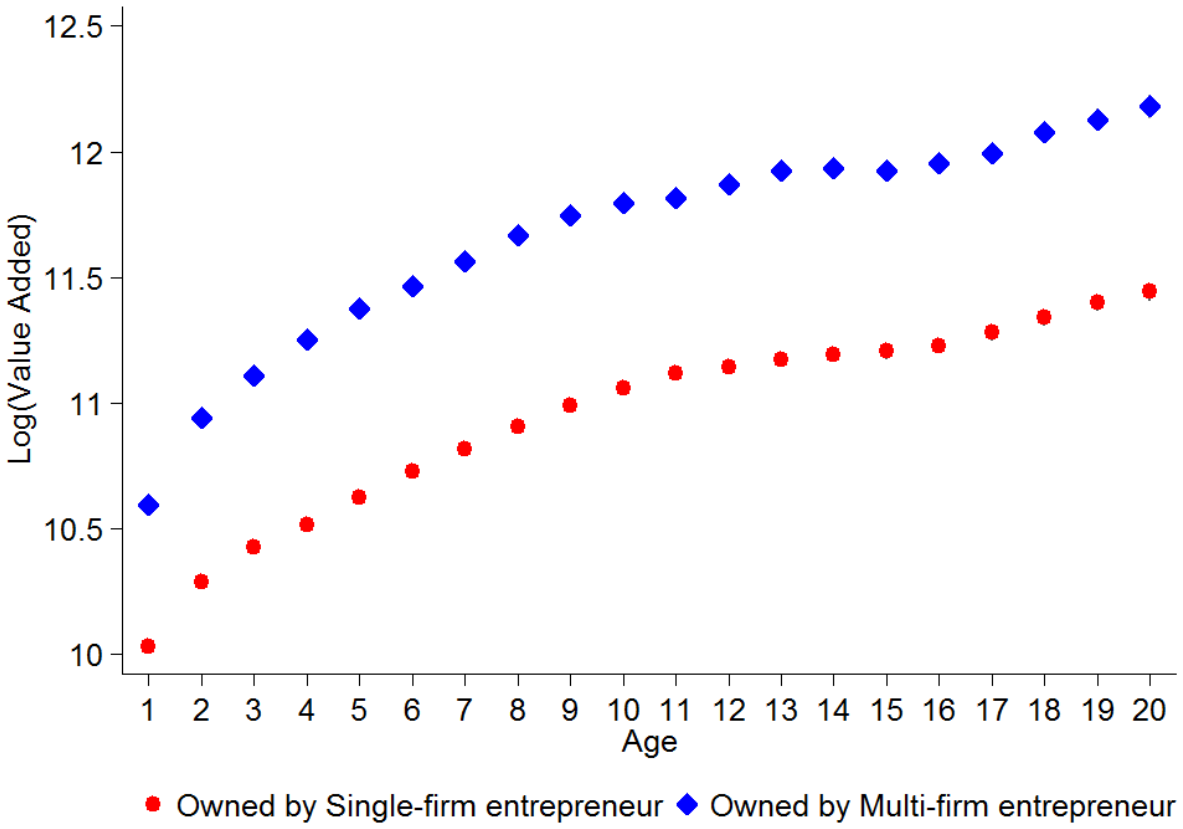
- Firm-specific productivity has persistence

Conclusion

- Using the unique feature of Thai names, this paper identifies entrepreneurs and the number of firms they own
- Multi-firm ownership is new channel which helps entrepreneurs overcome financial frictions
- Introduce a dynamic model with entrepreneurs that endogenously decides how many firms they want to own, weighing the benefits and costs.
- Calibrated model shows that the effect of financial frictions on the aggregate economy can be reduced by 1/3.

Appendix

Life cycle of firm size: raw data



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Regression specification

$$(y_{ijt}) = \gamma_i + \alpha_j + M_j + \sum_{a=2}^{20} \gamma_a D_{it}^a + \sum_{a=2}^{20} \beta_a D_{it}^a \times M_j + \epsilon_{ijt}$$

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Fact 2a: Firms own by multi-firm entrepreneurs grow faster

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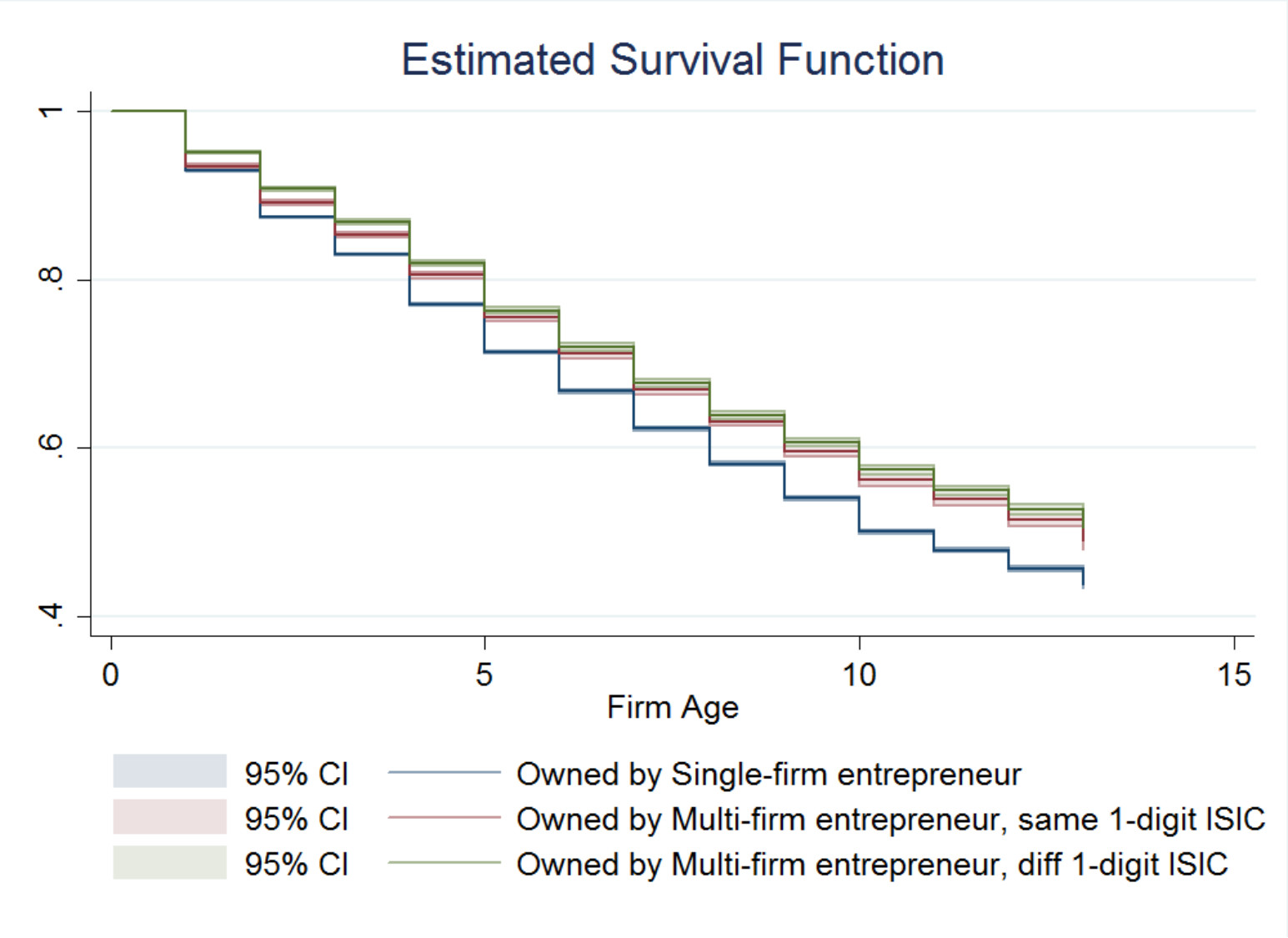
	(1) $\frac{k_{2015} - k_{2007}}{k_{2007}}$	(2) $\frac{k_{2015} - k_{2007}}{k_{2007}}$	(3) $\frac{k_{2015} - k_{2007}}{k_{2007}}$
Ln(Equity)	0.240***	0.254***	0.282***
	(0.012)	(0.012)	(0.019)
Return on Assets	0.490***	0.408***	0.571***
	(0.091)	(0.090)	(0.135)
Leverage	0.020***	0.211***	0.027***
	(0.002)	(0.002)	(0.003)
Total No. of Firms Owned by Entrepreneur			0.018*
			(0.009)
Ln(Capital)	-0.365***	-0.394***	-0.417***
	(0.008)	(0.009)	(0.013)
Firm-Year Obs.	76,228	76,228	37,012
R^2	0.0435	0.0535	0.0555
Sector FE	N	Y	Y
Geographical Location FE	N	Y	Y

Standard errors are in parentheses.

Firm survival function

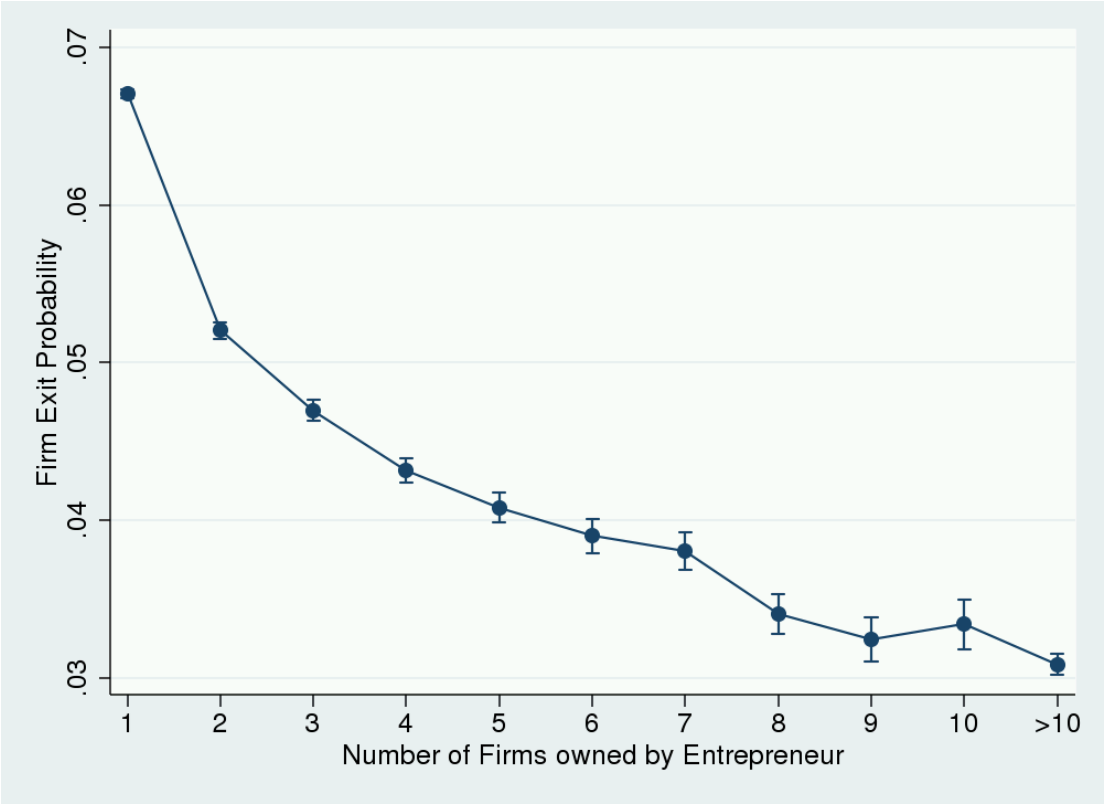
Go Back



Firm exits when it stops operating

Annual Firm Exit Rate and Entrepreneur owned firms

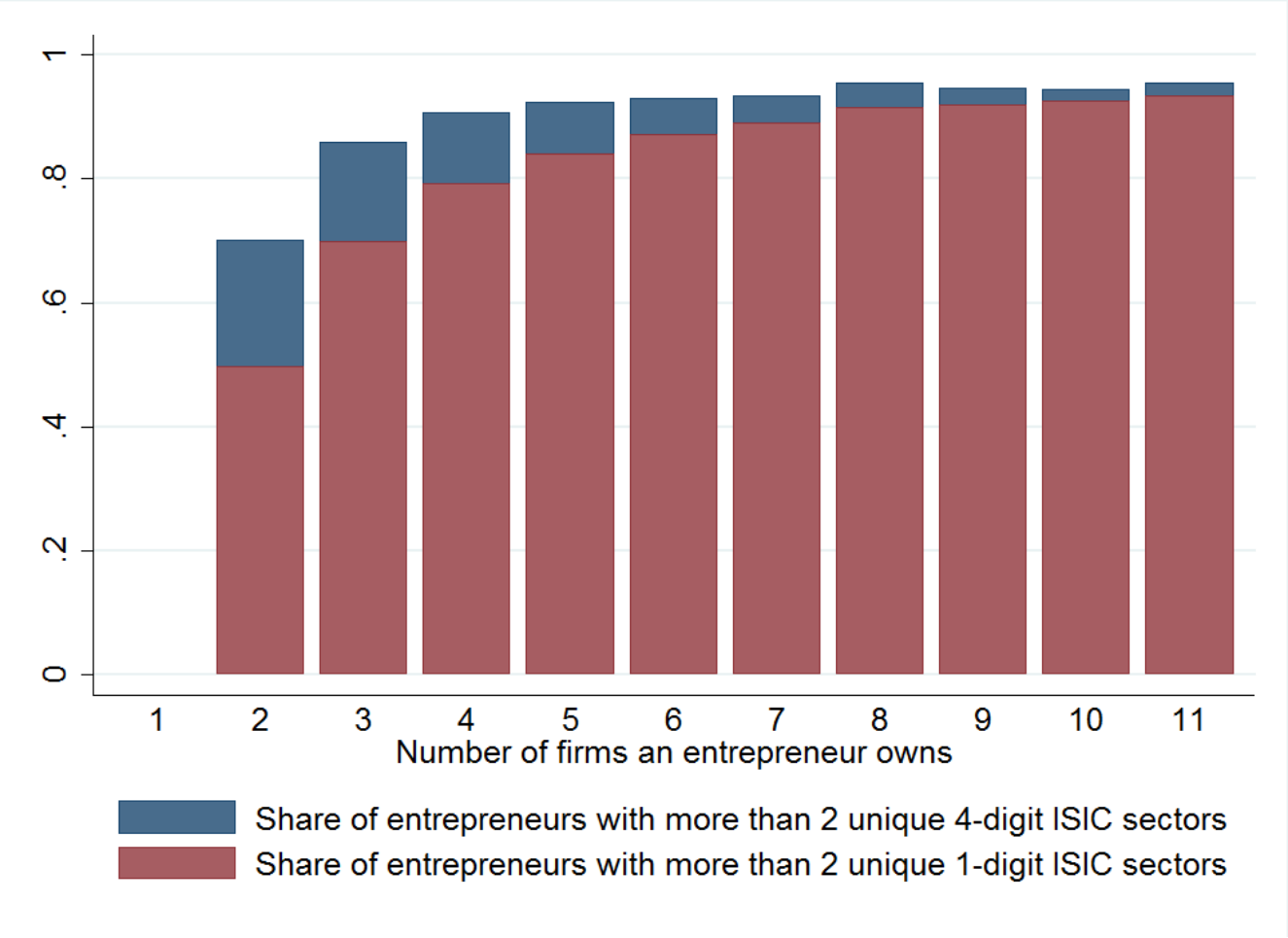
Go Back



I run a probit regression by regressing firm's exit rates on the number of firms an entrepreneur own, controlling for firm's age, firm's cohort, firm's 3-digit ISIC code and entrepreneur's experience. The firm exit rate is decreasing in the number of firms owned by entrepreneur.

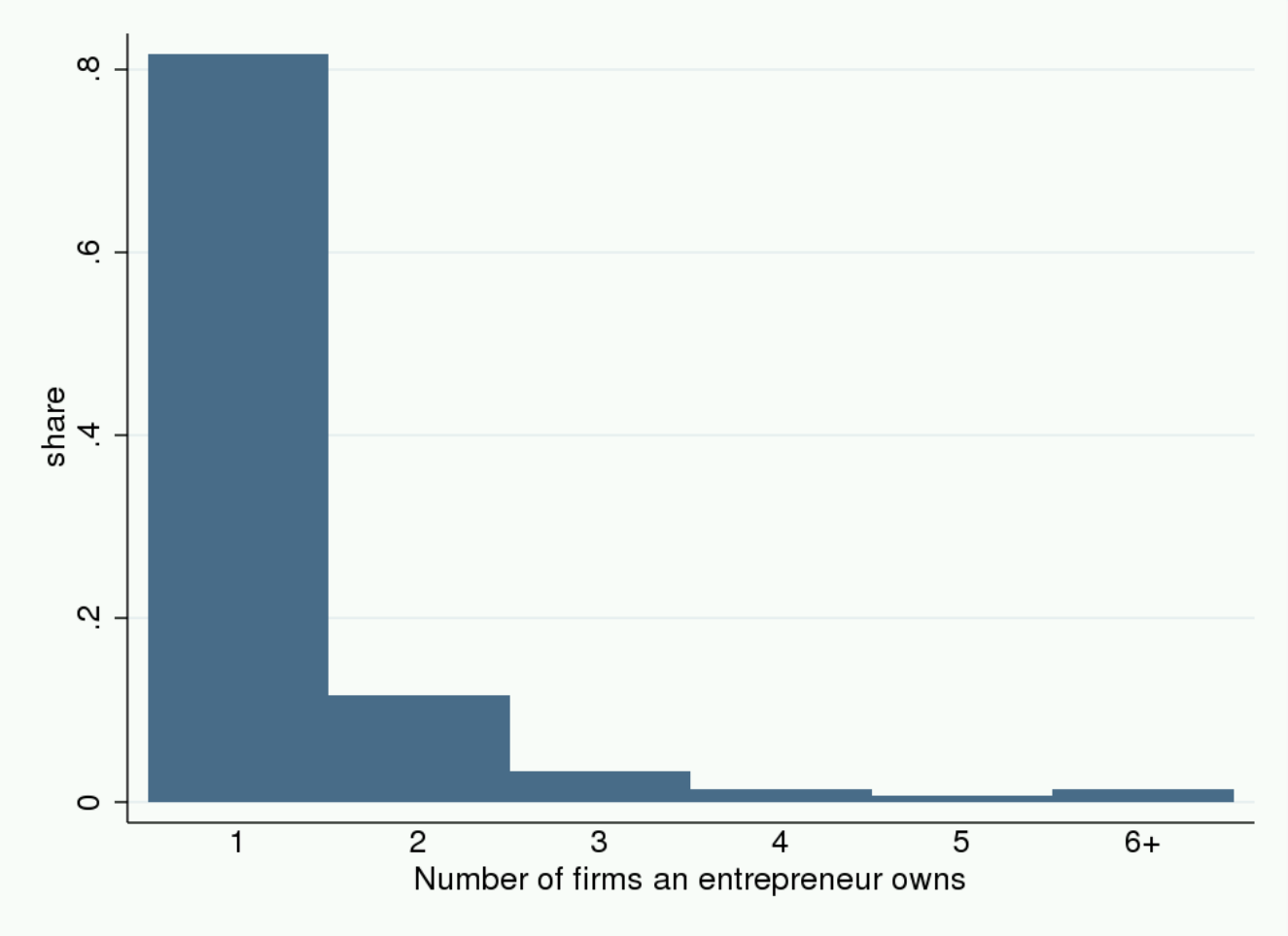
Entrepreneurs more likely to expand to other sectors as they have more firms

Go back



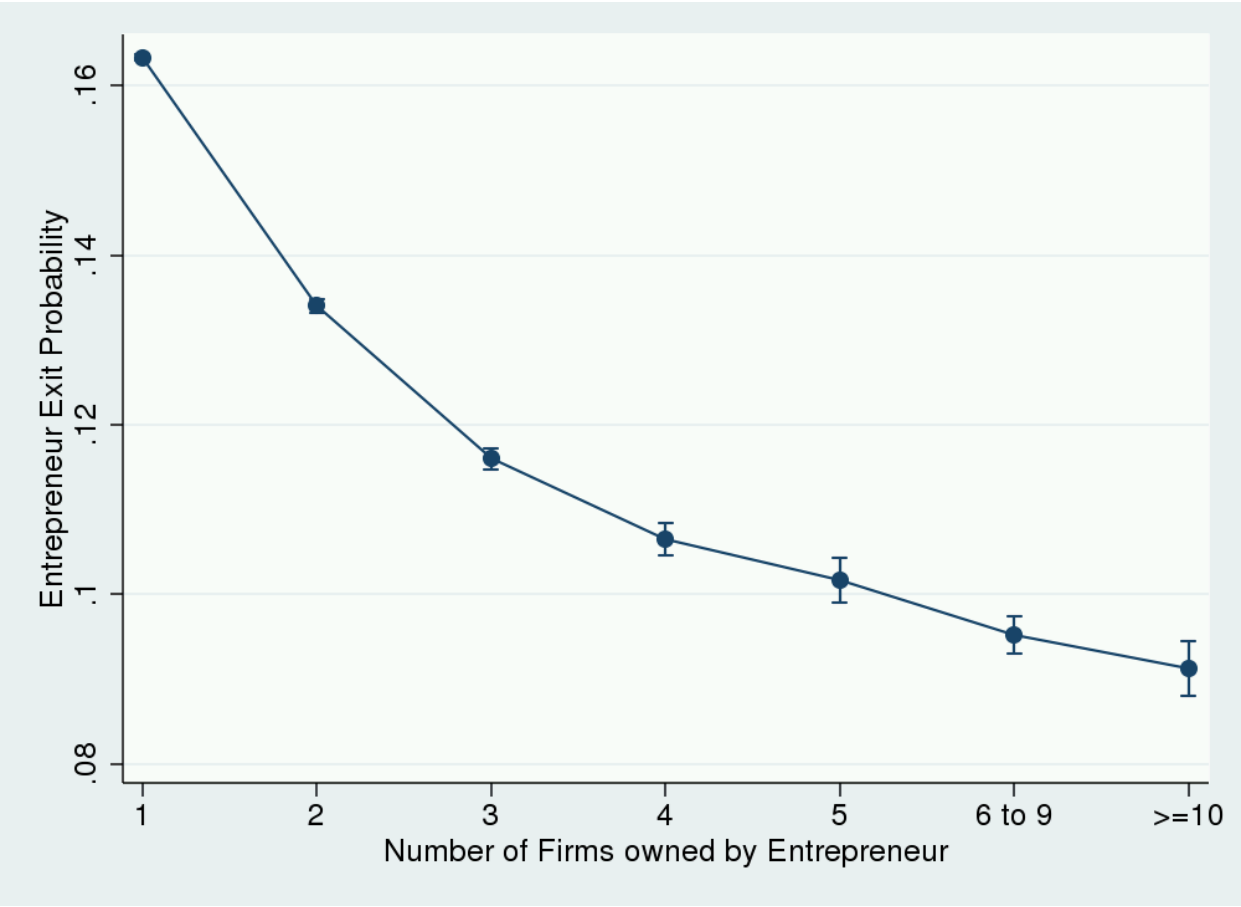
Share of entrepreneur with multiple firms

Go back



Entrepreneur exit rate (in 5 years)

$$Pr(D^5_{jt}{=}1) = \Phi\left(\gamma_0 + \beta(\text{No. of Firms})_{jt} + \gamma_1 \text{Exp}_{jt} + \text{Industry}_j + \epsilon_{jt}\right)$$



Entrepreneur "exits" when he no longer has any operating firms.

Person Name Act (1962)

Section 8.2:

> *No last name may be identical or intended to be similar to any title, save where the title belongs to the person in question himself, his ascendant or his descendant.*

Go Back

Data Variables

- Firm variables
  - Registry date
  - Years of financial statement submission
  - Operating status
  - Industry sector
  - Board members' names
- Balance sheet variables (2007 to 2015)
  - Capital
  - Current and non-current assets, liability, equity
- Income statement variables (2007 to 2015)
  - Total, main revenue
  - Total cost of production
  - Interest paid
  - Administrative expenses
  - Taxes
  - Profits

Back

Multi-firm entrepreneurs accumulate more wealth – over whole sample period

Back

	(1) $\frac{W_{2015} - W_{2007}}{W_{2007}}$	(2) $\frac{W_{2015} - W_{2007}}{W_{2007}}$	(3) $\frac{W_{2015} - W_{2007}}{W_{2007}}$
Ln(Wealth)	-0.551***	-0.555***	-0.794***
	(0.015)	(0.015)	(0.027)
Total No. of Firms Owned by Entrepreneur	0.260***	0.295***	0.021
	(0.014)	(0.014)	(0.018)
Experience	-0.016***	-0.038***	-0.072***
	(0.003)	(0.003)	(0.007)
Experience^2	0.0002**	0.0006***	0.001***
	(0.00007)	(0.00007)	(0.0001)
Entrepreneur-Year Obs.	34,439	34,439	9,840
R^2	0.0857	0.1275	0.2480
First Sector FE	N	Y	Y
Sector Interaction FE	N	N	Y

Standard errors are in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Approximating $\sum_{j=1}^{N} \exp(e_j)$ Back

- Sum of log normal variables can be approximated by a log normal variable.
- $\sum_{j=1}^{N} \exp(e_j) \approx \exp(\hat{e})$
- $\hat{e} \sim \mathcal{N}\left(\hat{\mu}(N,\rho), \hat{\sigma}^2(N,\rho)\right)$ where

$$\begin{aligned} \hat{\sigma}^2(N,\rho) &= \ln\left[\frac{\exp(\sigma_e^2) + (N-1)\exp(\rho\sigma_e^2)}{N}\right] \\ \hat{\mu}(N,\rho) &= \ln(N) + \frac{\sigma_e^2}{2} - \frac{\hat{\sigma}^2(N,\rho)}{2} \end{aligned} \tag{10}$$

- $\tilde{\pi}(z, a, \vec{e}, N) \approx \pi(z, a, \hat{e}, N)$

Gains in having multiple firms: scale and volatility

- Under frictionless world, $\pi^P(z, \hat{e}, \rho, N)$ follows a log normal distribution with $\hat{\mu}(N, \rho)$ and $\hat{\sigma}^2(N, \rho)$.
- $\pi^P(z, \hat{e}, \rho, N) = (1 - \alpha) \left[\left(\frac{\alpha}{R} \right)^{\alpha} \right]^{\frac{1}{1-\alpha}} \exp(z + \hat{e})$
- **Scale**: Compare $\pi^P(z, \hat{e}, \rho, N)$ to $\pi^P(z, \hat{e}, \rho, N + 1)$
- **Volatility**: Compare $\pi^P(z, \hat{e}, \rho_1, N)$ to $\pi^P(z, \hat{e}, \rho_2, N)$ for $\rho_1 \leq \rho_2$.

Initial wealth affects borrowing

Go Back

$$\frac{L_{2014} - L_{2007}}{L_{2007}} = d_0 + \alpha \log(A_{2007}) + \gamma \log(k_{2007}) + \beta \log(Z_{2007}) + u_t$$

	(1) $\frac{L_{2014} - L_{2007}}{L_{2007}}$	(2) $\frac{L_{2014} - L_{2007}}{L_{2007}}$	(3) $\frac{L_{2014} - L_{2007}}{L_{2007}}$
Ln(Capital)	-1.13***	-1.16***	-2.33***
	(0.14)	(0.15)	(0.16)
Ln(Equity)	0.597***	0.553***	0.938***
	(0.13)	(0.13)	(0.24)
Ln(Productivity)	0.241**	0.354***	
	(0.12)	(0.13)	
Total No. of Firms owned by Entrepreneur			0.698**
			(0.32)
Firm-Year Obs.	4240	4240	4233
Sector FE	N	Y	Y
Geographical Location FE	N	Y	Y

Approximating $\sum_{j=1}^{N} \exp(e_j)$ Go Back

- Sum of log normal variables can be approximated by a log normal variable.
- $\sum_{j=1}^{N} \exp(e_j) \approx \exp(\hat{e})$
- $\hat{e} \sim \mathcal{N}\left(\hat{\mu}(N,\rho), \hat{\sigma}^2(N,\rho)\right)$ where

$$\begin{aligned} \hat{\sigma}^2(N,\rho) &= \ln\left[\frac{\exp(\sigma_e^2) + (N-1)\exp(\rho\sigma_e^2)}{N}\right] \\ \hat{\mu}(N,\rho) &= \ln(N) + \frac{\sigma_e^2}{2} - \frac{\hat{\sigma}^2(N,\rho)}{2} \end{aligned} \tag{11}$$

- $\tilde{\pi}(z, a, \vec{e}, N) \approx \pi(z, a, \hat{e}, N)$

Gains in having multiple firms: scale

$\pi^P(z, \hat{e}, \rho, N+1)$ **second-order stochastically dominates (SOSD)** $\pi^P(z, \hat{e}, \rho, N)$

$$\begin{aligned} \hat{\sigma}^2(N+1, \rho) &\leq \hat{\sigma}^2(N, \rho) \\ \hat{\mu}(N+1, \rho) &> \hat{\mu}(N, \rho) \\ E\left[\pi^P(z, \hat{e}, \rho, N+1)\right] &> E\left[\pi^P(z, \hat{e}, \rho, N)\right] \end{aligned} \tag{12}$$

$\Rightarrow$

$$E\left[U(\pi^P(z, \hat{e}, \rho, N+1))\right] > E\left[U(\pi^P(z, \hat{e}, \rho, N))\right]$$

for any concave and weakly increasing function..

Gains in having multiple firms: volatility

For any $\rho_2 > \rho_1$, $\pi^P(z, \hat{e}, \rho_2, N)$ **is a mean-preserving spread of** $\pi^P(z, \hat{e}, \rho_1, N)$

$\Rightarrow \pi^P(z, \hat{e}, \rho_1, N)$ **SOSD** $\pi^P(z, \hat{e}, \rho_2, N)$

$$\begin{aligned} \hat{\sigma}^2(N, \rho_1) &< \hat{\sigma}^2(N, \rho_2) \\ \hat{\mu}(N, \rho_1) &> \hat{\mu}(N, \rho_2) \\ E\left[\pi^P(z, \hat{e}, \rho_1, N)\right] &= E\left[\pi^P(z, \hat{e}, \rho_2, N)\right] \end{aligned} \tag{13}$$

$\Rightarrow$

$$E\left[U(\pi^P(z, \hat{e}, \rho_1, N))\right] > E\left[U(\pi^P(z, \hat{e}, \rho_2, N))\right]$$